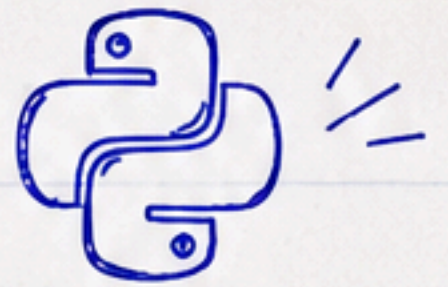


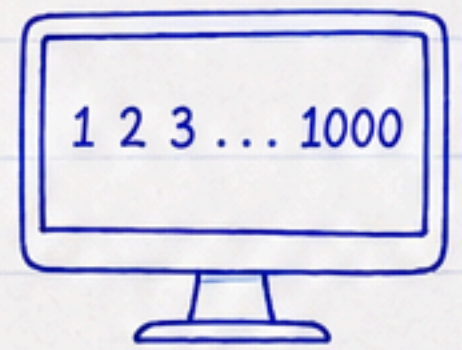


CHAPTER 07

Loops In Python



Sometimes we want to repeat a set of statements in our program. For instance: **Print 1 to 1000.**

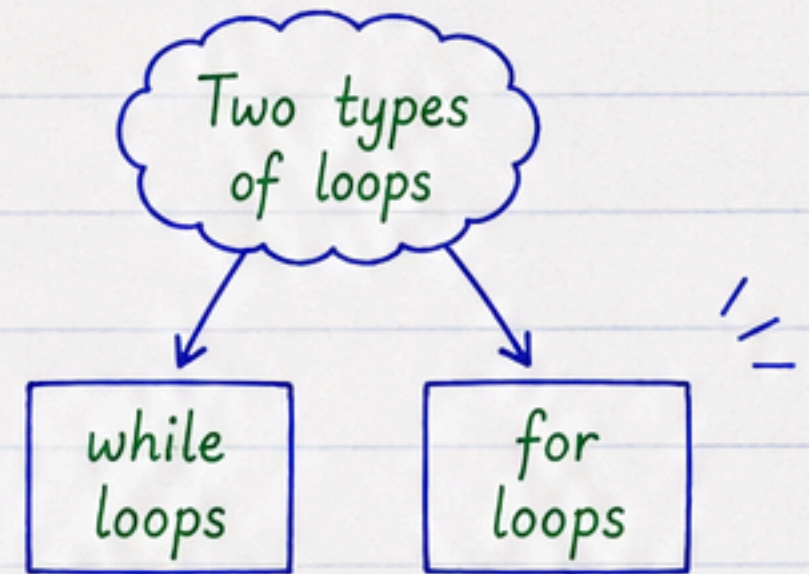


Loops make it easy for a programmer to tell the computer which set of instructions to repeat and how!

Types Of Loops In Python

Primarily there are two types of loops in python.

- while loops
- for loops



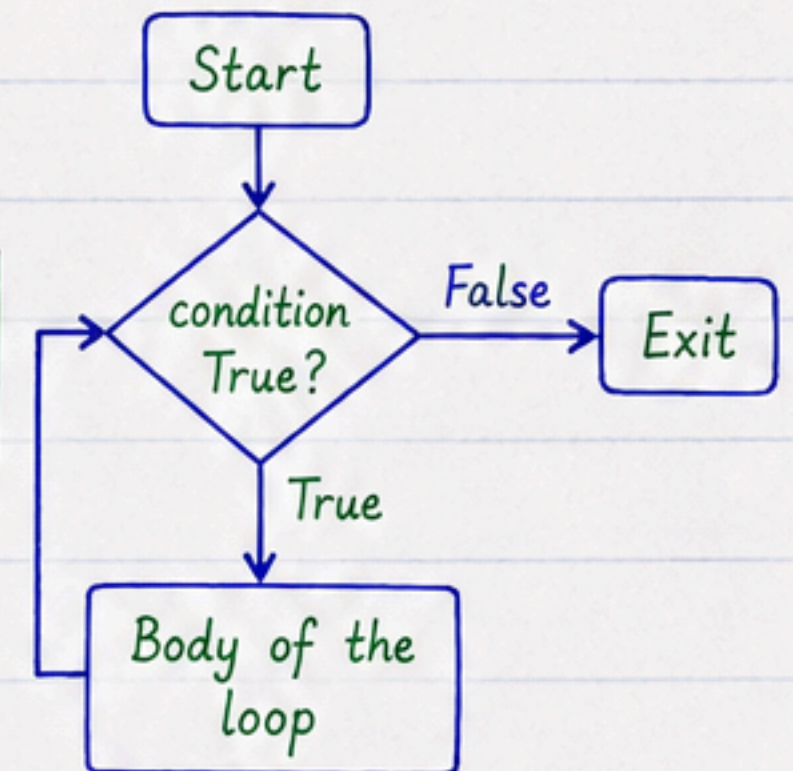
We will look into these one by one.

While Loop

Syntax:

```
while (condition): # The block keeps executing until the condition is true
    # Body of the loop
```

- In while loops, the condition is checked first. If it evaluates to **true**, the body of the loop is executed otherwise not!
- If the loop is entered, the process of **[condition check & execution]** is continued until the condition becomes **False**.



Quick Quiz: Write a program to print 1 to 50 using a while loop.

Example:

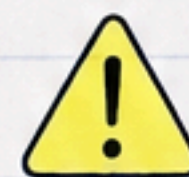
```
i = 0
while i < 5: # print "Harry" - 5 times!
    print("Harry")
    i = i + 1
```

Output

Harry
Harry
Harry
Harry
Harry



Note: If the condition never become false, the loop keeps getting executed.



Quick Quiz: Write a program to print the content of a list using while loops.



For Loop

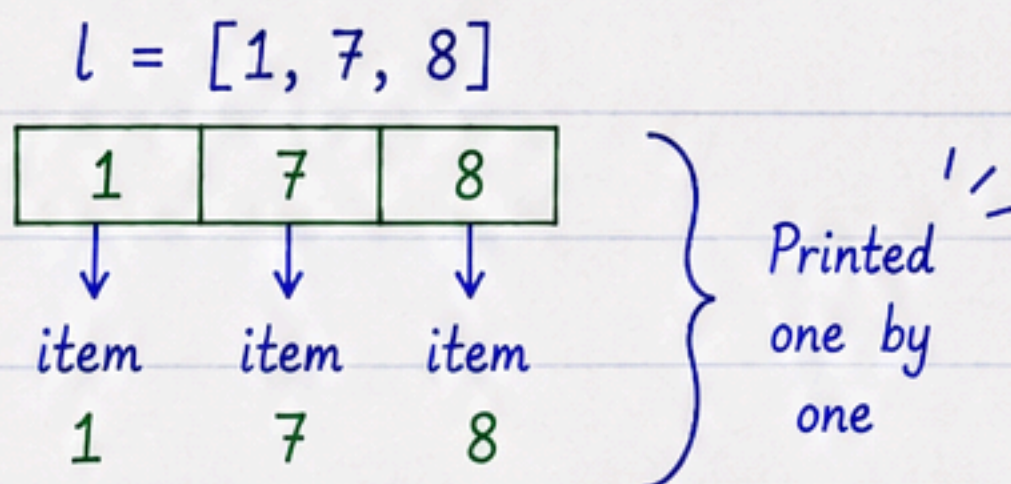
A for loop is used to iterate through a sequence like list, tuple, or string [iterables]

Iterate →
to repeat one by one



Syntax:

```
l = [1, 7, 8]
for item in l:
    print(item)    # prints 1, 7 and 8
```



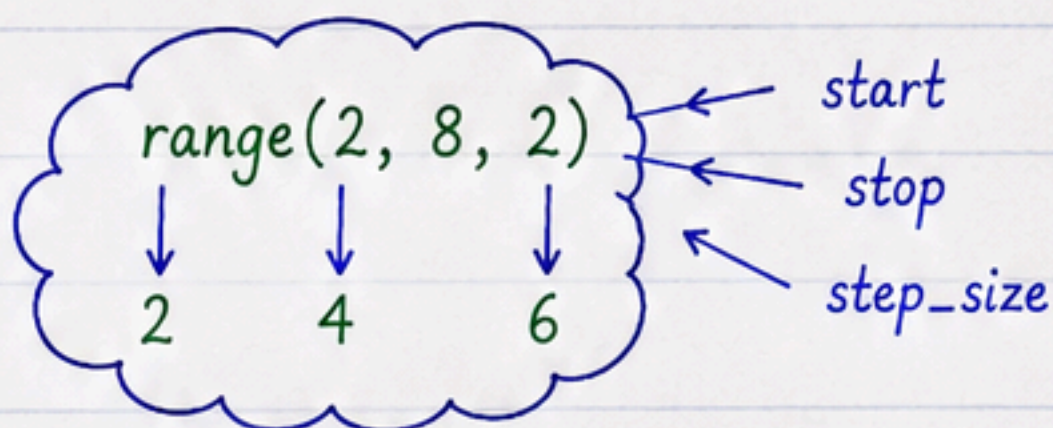
Range Function In Python

The range() function in python is used to generate a sequence of number.

We can also specify the start, stop and step-size as follows:

```
range(start, stop, step_size)
```

step_size is usually not used with range()



Remember:

- start → included
- stop → excluded
- step_size → how many numbers to skip



An Example Demonstrating Range() Function.

```
for i in range(0,7): # range(7) can also be used
    print(i)         # prints 0 to 6
```

Output:

0
1
2
3
4
5
6

range(0,7)
produces
numbers from
0 to 6

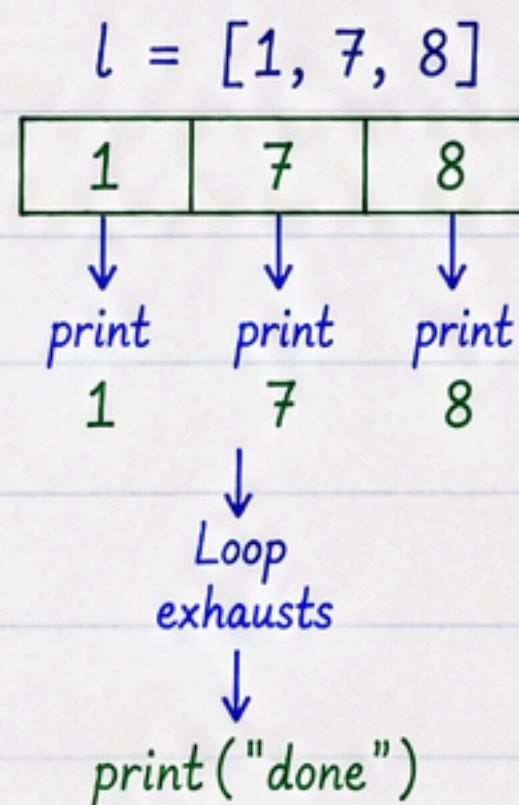


For Loop With Else

An optional else can be used with a for loop if the code is to be executed when the loops exhausts.

Example:

```
l = [1, 7, 8]
for item in l:
    print(item)
else:
    print("done") # this is printed when the loop exhausts!
```



Else block runs only if the for loop completes normally.

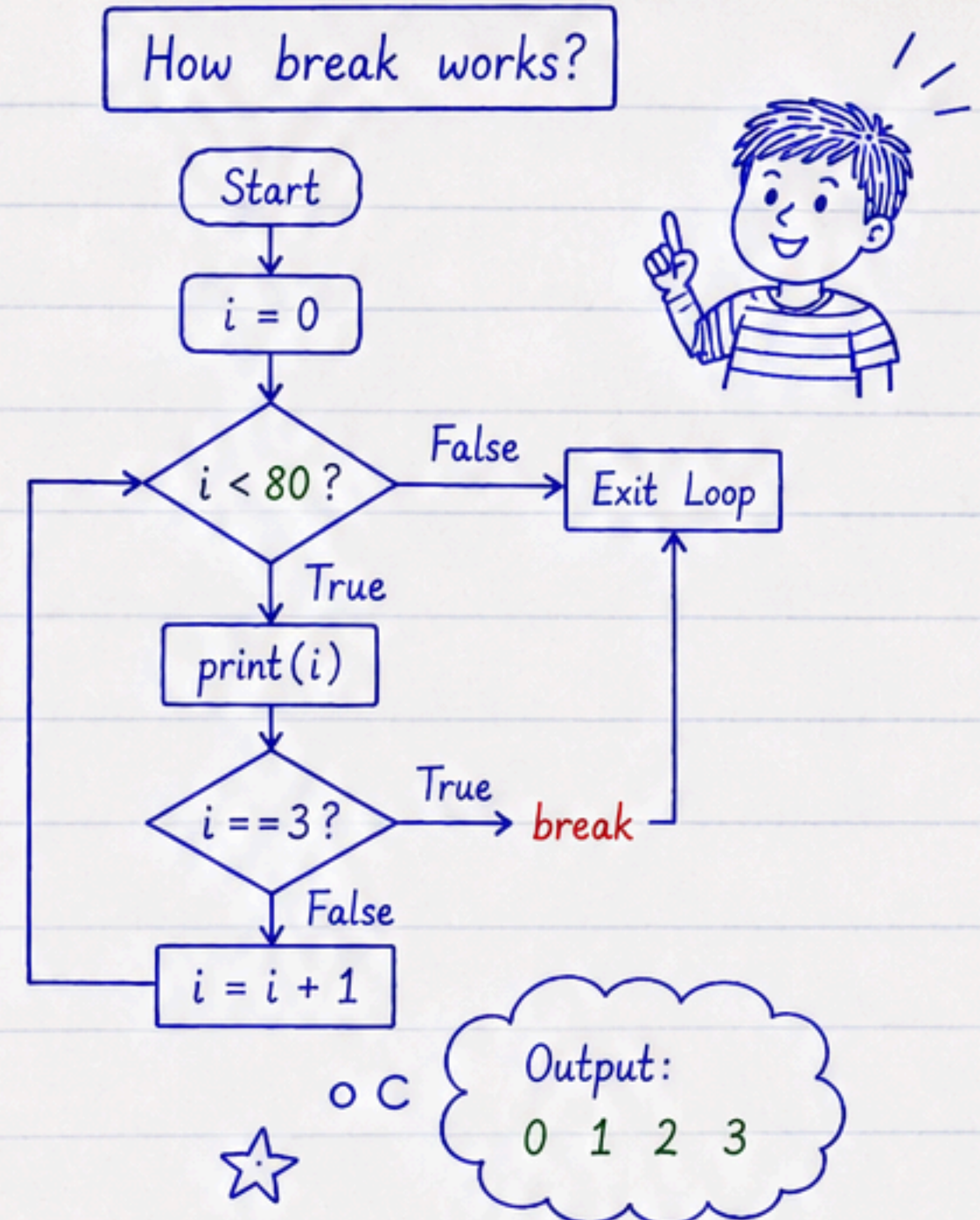


The Break Statement

'break' is used to come out of the loop when encountered. It instructs the program to - exit the loop now.

Example:

```
for i in range(0,80):  
    print(i)    # this will print 0,1,2 and 3  
    if i == 3:  
        break
```



The Continue Statement

'continue' is used to stop the current iteration of the loop and continue with the next one. It instructs the Program to "skip this iteration".

Example:

```
for i in range(4):  
    print("printing")  
    if i == 2: # if i is 2, the iteration is skipped  
        continue  
    print(i)
```

i	print("printing")	if i==2	print(i)
0	printing	False	0
1	printing	False	1
2	printing	True	Skipped
3	printing	False	3

Output:
printing
0
printing
1
printing
3

When i is 2, continue skips the rest of the code in this iteration.

Pass Statement

pass is a null statement in python. It instructs to "do nothing".

Example:

```
l = [1,7,8]  
for item in l:  
    pass # without pass, the program  
        will throw an error
```

Why do we need pass?

Python does not allow empty blocks. We use pass as a placeholder when we don't want to write any code yet.

Without pass (Error)

```
l = [1,7,8]  
for item in l:  
    # IndentationError: expected  
    an indented block
```



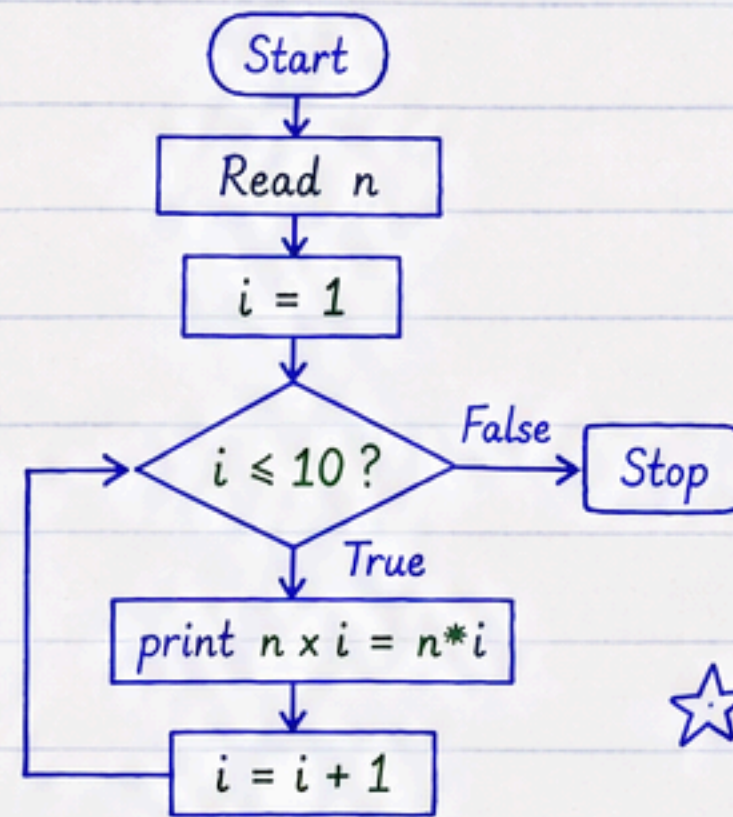


Practice Set



1. Write a program to print multiplication table of a given number using for loop.

```
n = int(input("Enter a number: "))
print("Multiplication table of", n)
for i in range(1, 11):
    print(n, "x", i, "=", n*i)
```



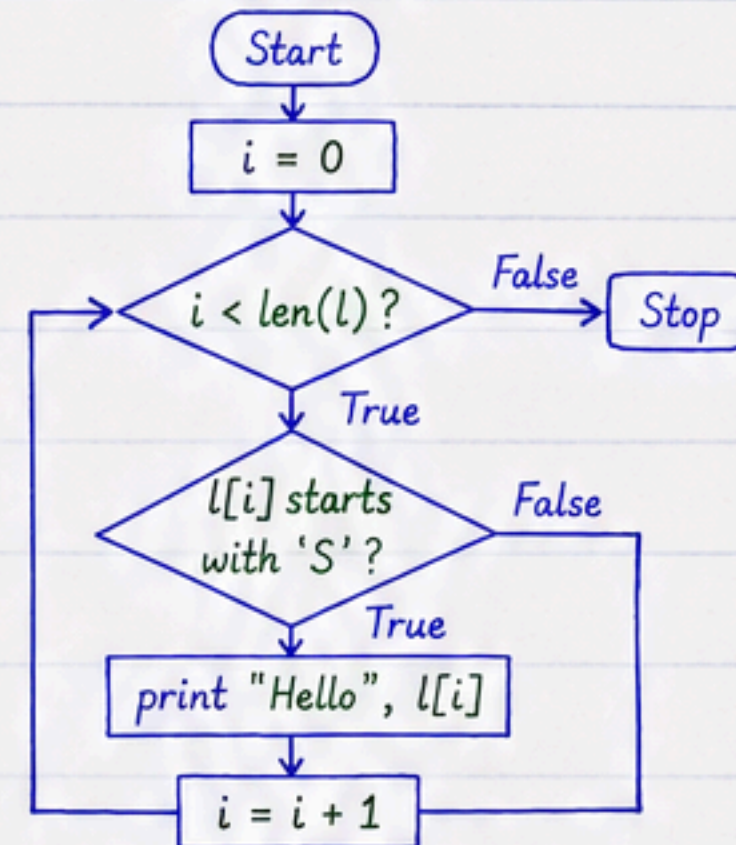
Example:
n = 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50



2. Write a program to greet all the person names stored in a list 'l' and which starts with S.

```
l = ["Harry", "Soham", "Sachin", "Rahul"]
```

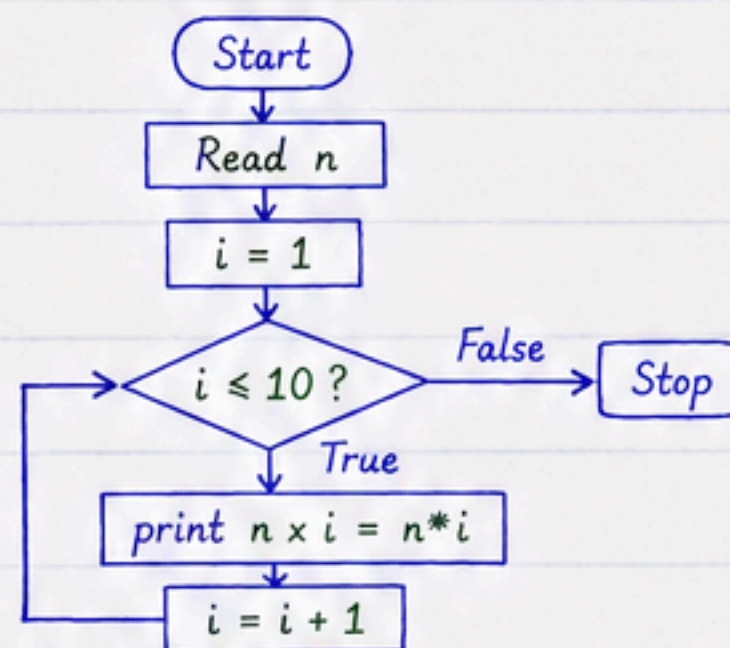
```
l = ["Harry", "Soham", "Sachin", "Rahul"]
for name in l:
    if name.startswith('S'):
        print("Hello", name)
```



Output:
Hello Soham
Hello Sachin

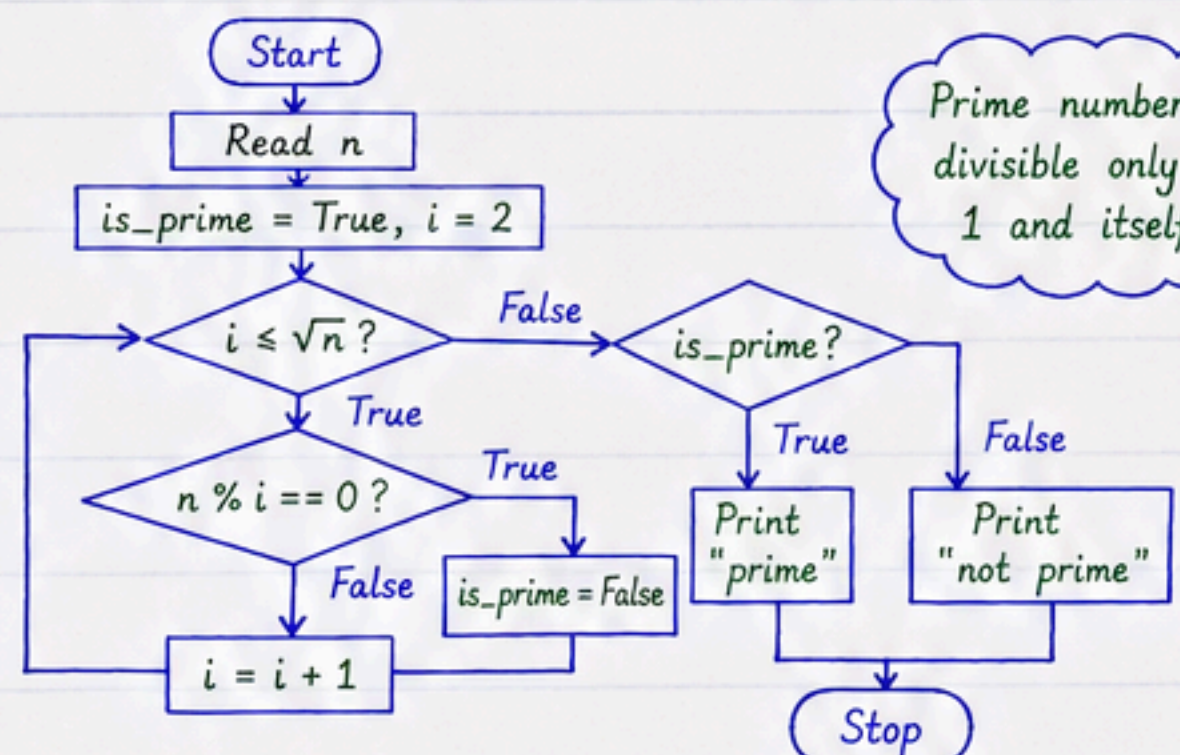
3. Attempt problem 1 using while loop.

```
n = int(input("Enter a number: "))
i = 1
print("Multiplication table of", n)
while i ≤ 10:
    print(n, "x", i, "=", n*i)
    i = i + 1
```



4. Write a program to find whether a given number is prime or not.

```
n = int(input("Enter a number: "))
is_prime = True
if n < 2:
    is_prime = False
else:
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            is_prime = False
            break
if is_prime:
    print(n, "is prime")
else:
    print(n, "is not prime")
```

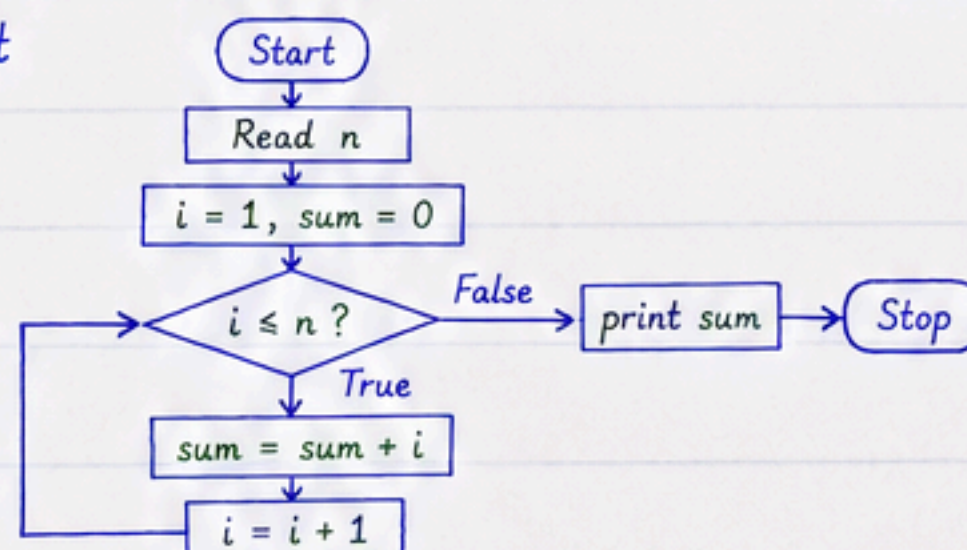


Prime number is divisible only by 1 and itself.



5. Write a program to find the sum of first n natural numbers using while loop.

```
n = int(input("Enter n: "))
i = 1
sum = 0
while i ≤ n:
    sum = sum + i
    i = i + 1
print("Sum =", sum)
```



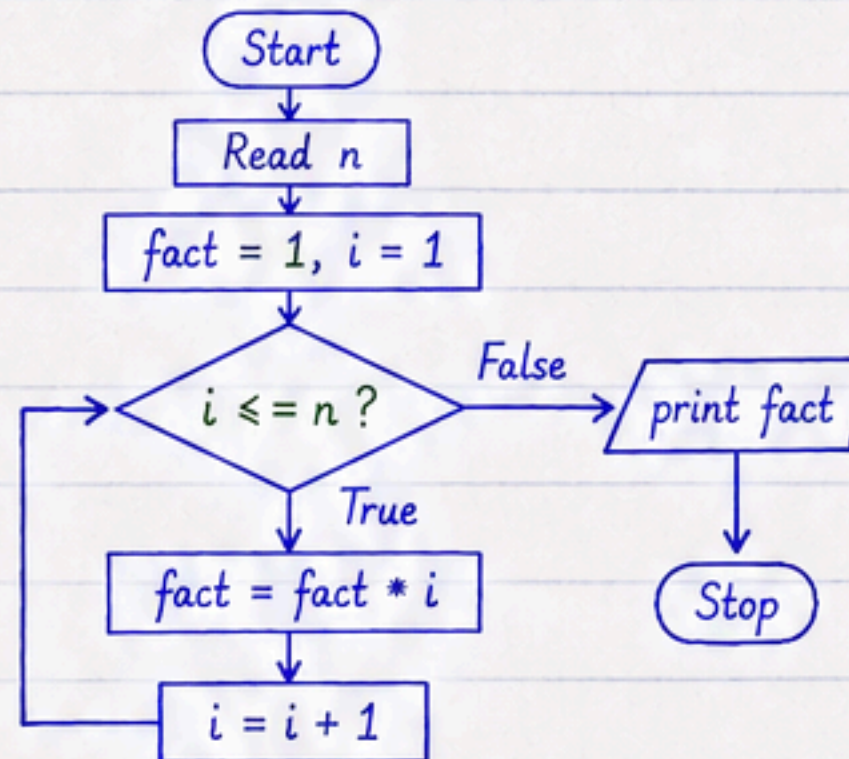


Practice Set (Continued)



6. Write a program to calculate the factorial of a given number using for loop.

```
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n+1):
    fact = fact * i
print("Factorial of", n, "is", fact)
```



Example:
n = 5
Output:
Factorial of
5 is 120

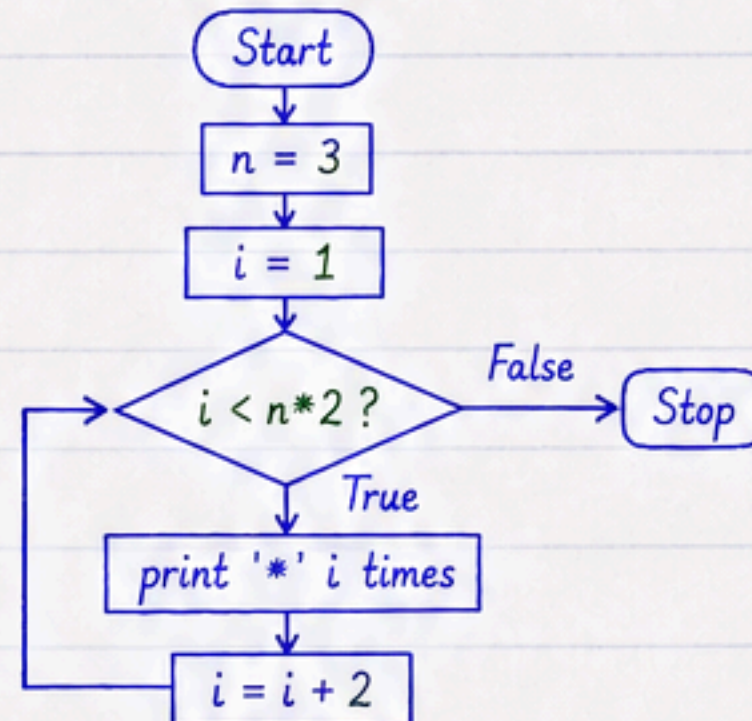


7. Write a program to print the following star pattern.

for n = 3

```
*
***
*****
```

```
n = 3
for i in range(1, n*2, 2):
    for j in range(i):
        print("*", end=" ")
    print()
```



Output:
*

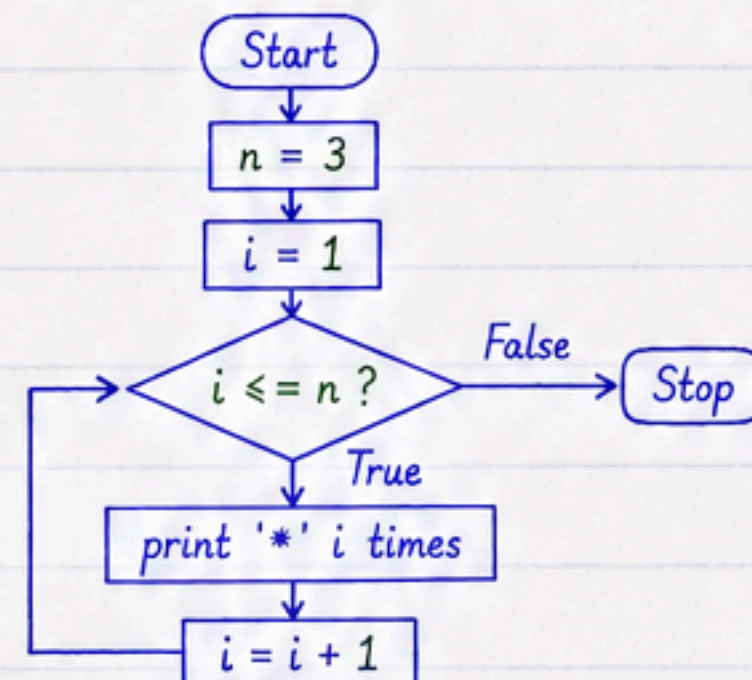


8. Write a program to print the following star pattern.

for n = 3

```
*
**
***
```

```
n = 3
for i in range(1, n+1):
    for j in range(i):
        print("*", end=" ")
    print()
```



Output:
*
**

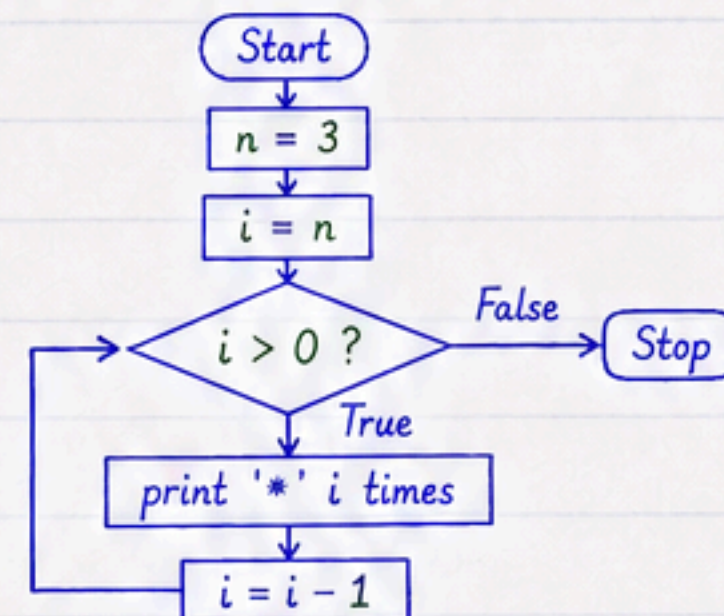


9. Write a program to print the following star pattern.

for n = 3

```
* * *
* *
* * *
```

```
n = 3
for i in range(n, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

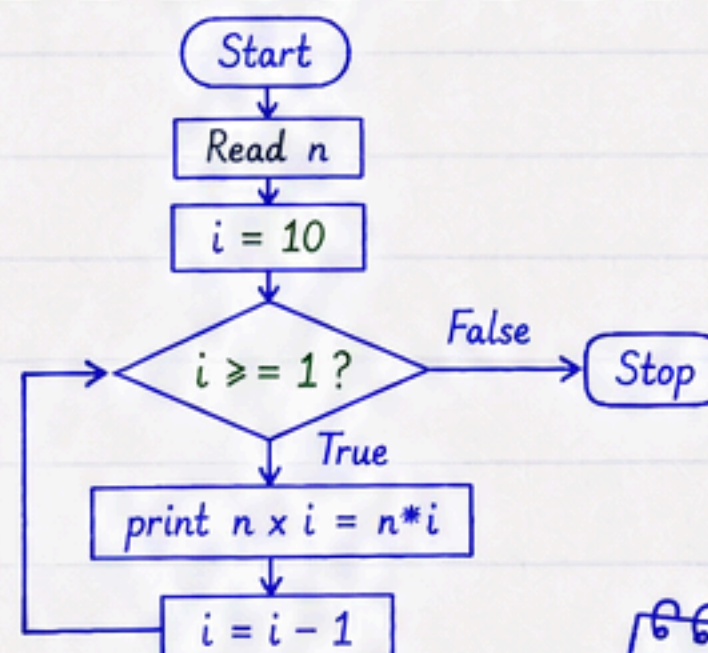


Output:
* * *
* *
* * *



10. Write a program to print multiplication table of n using for loops in reversed order.

```
n = int(input("Enter a number: "))
for i in range(10, 0, -1):
    print(n, "x", i, "=", n*i)
```



Example:
n = 5
Output:
5 x 10 = 50
5 x 9 = 45
5 x 8 = 40
5 x 7 = 35
5 x 6 = 30
5 x 5 = 25
5 x 4 = 20
5 x 3 = 15
5 x 2 = 10
5 x 1 = 5

